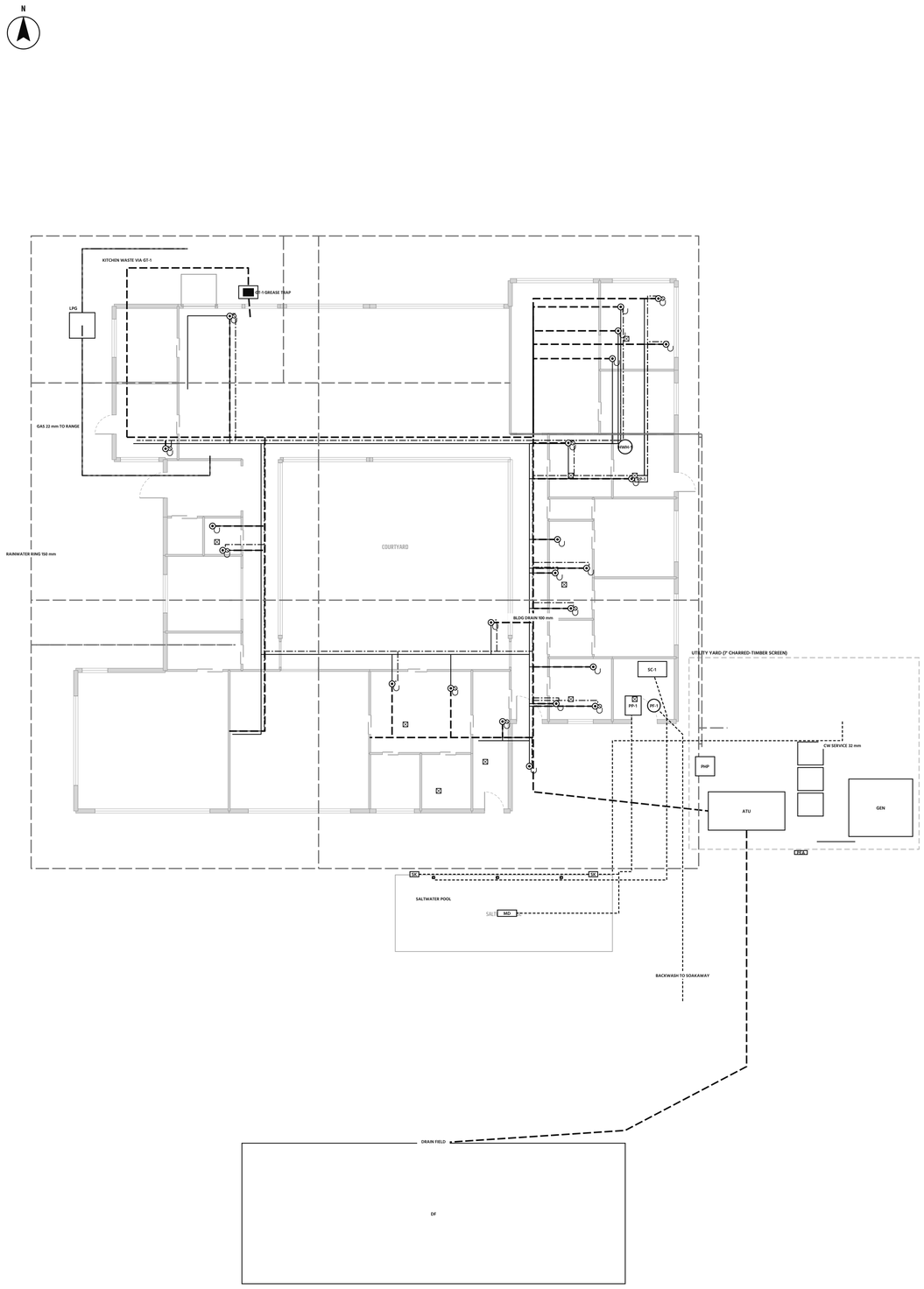

1 **PLUMBING + SITE UTILITIES PLAN**

SCALE: 3/32" = 1'-0"

PLUMBING LEGEND

- Domestic cold water (PPR)
- Domestic hot water + recirculation
- Sanitary drainage (uPVC, under slab)
- Vent stack through roof
- Rainwater leader + collector
- Pool suction / return / fill
- Plumbing fixture

FIXTURE SCHEDULE

FIXTURE	CONNECTION	QTY
Cold Plunge (Chiller + Filter)	Auto-fill + 50 mm drain; P-trap at fixture, vented	1
Courtyard Shower	Cold + hot, to rain garden; P-trap at fixture, vented	1
Dog Wash Basin	50 mm drain with hair trap; P-trap at fixture, vented	1
Hinoki Soaking Tub	50 mm drain, 20 mm fill; P-trap at fixture, vented	1
Integrated Quartzite Sink + Dishwasher + Kitchen RO Tap	50 mm drain to grease trap; RO-1 under sink; P-trap at fixture, vented	1
Lavatory	40 mm drain, 15 mm supplies; P-trap at fixture, vented	5
Outdoor Shower	Cold + hot, to rain garden; P-trap at fixture, vented	1
Shower (Linear Drain)	50 mm drain, thermostatic valve; P-trap at fixture, vented	4
Steam Generator Feed	15 mm cold + drain; P-trap at fixture, vented	1
Tub-Shower	50 mm drain; P-trap at fixture, vented	1
Utility Sink	40 mm drain; P-trap at fixture, vented	1
Washer Box	50 mm standpipe; P-trap at fixture, vented	1
Water Closet (Wall-Hung, Dual Flush)	3/6 L, 100 mm drain; P-trap at fixture, vented	4

EQUIPMENT SCHEDULE

TAG	EQUIPMENT	SPECIFICATION
HWH-1	Heat-Pump Water Heater HWH-1	300 L, 1.5 kW heat pump with 3 kW backup element
RP-1	Recirculation Pump RP-1 + Manifold	Timer + thermostat control, 15 mm
GT-1	Grease Interceptor GT-1	100 L, precast, lid at grade
PP-1	Pool Pump (Variable Speed)	1.5 kW, VS pump, 50 mm suction / return
PF-1	Pool Media Filter	600 mm glass-media filter with 6-way valve
SC-1	Salt Chlorine Generator Cell + Controller	Sized for 60 m³, 1,000–4,000 ppm
PHP	Pool Heat Pump (Optional)	13 kW, in utility yard
RI	Pool Return Inlet	40 mm
RI	Pool Return Inlet	40 mm
RI	Pool Return Inlet	40 mm
SK	Skimmer	Wide-mouth
SK	Skimmer	Wide-mouth
MD	Main Drains (Dual, Anti-Entrapment)	2 x VGB-compliant covers
ATU	Septic Tank + Aerobic Treatment Unit	3,000 L septic + 2,000 L ATU (TIS 2550 type), lids at grade
DF	Drain Field (Leach Trenches)	60' x 22', 4 trenches, perc test required
GEN	Generator + Battery Pad	20 kVA diesel standby (auto-start) + 2 x 13.5 kWh LFP batteries, hybrid inverter
PEA	PEA Meter Pedestal + Main Isolator	3-phase 400 V, 300/5 A meter, surge protection

TRAPS, VENTS + GREASE

- Trapped floor drain FD (deep seal + trap primer)
- P-trap at every fixture
- Vent stack to open air above roof
- Grease interceptor GT-1 (kitchen, before septic)

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
3/32" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

PLUMBING NOTES

Supply: well + rainwater, treated (sediment, carbon, UV, softener), booster pump 3 bar.

Sanitary to septic + ATU (TIS 2550 effluent) and drain field; perc test required.

Saltwater backwash and pool drain go to a lined soakaway, never to septic or garden.

Kitchen through grease interceptor GT-1 (Thai Building Control Act).

KITCHEN GAS + WATER

LPG 2 x 48 kg cage, auto-changeover regulator, 22 mm line in sleeve to the 48" range (wok burner) with leak-interlocked solenoid. Optional 15 mm branch to the outdoor wok station.

Integrated quartzite sink: 50 mm waste to grease trap GT-1; RO-1 drinking tap + fridge ice line. See A-401.

STANDARD

PPR fusion-welded supply, pressure test 1.5 x WP (min 10 bar, 2 h).

Waterproofing membrane in all wet areas; central heat-pump hot water.

SHEET TITLE

Plumbing + Site Utilities Plan

SHEET

P-101